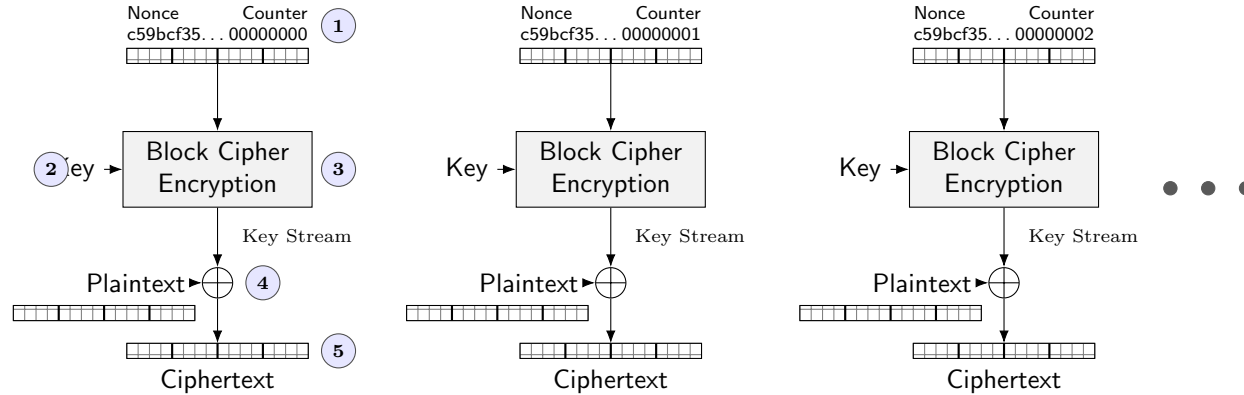
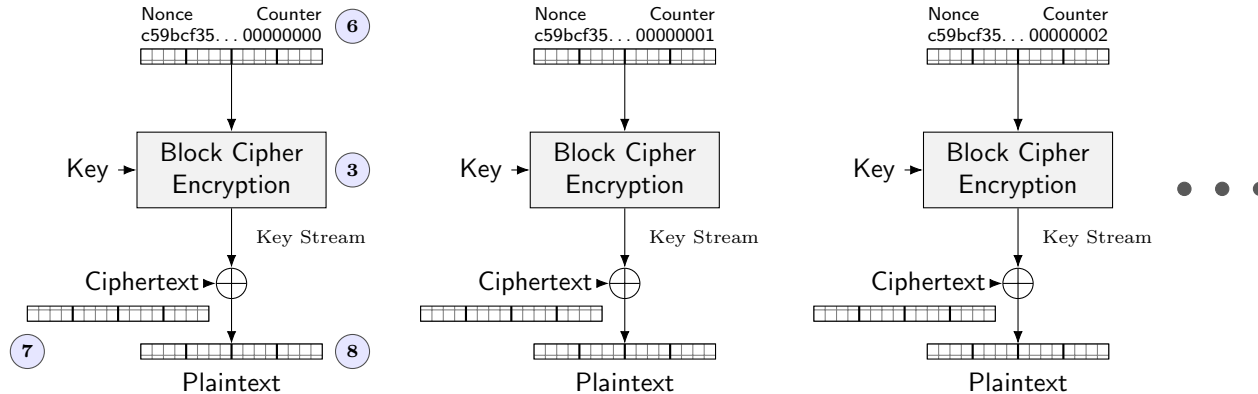


AES COUNTER (CTR) MODE: ENCRYPTION AND DECRYPTION

Encryption Process



Decryption Process



PROCESS EXPLANATIONS

Encryption (Top)

- Nonce & Counter:** Forms the static and incrementing part of the block cipher input.
- Key Input:** Used with symmetric Block Cipher to create a Key Stream.
- Key Stream Gen:** The block cipher encrypts the Nonce+Counter.
- Key Stream XOR:** Key stream is XORed with Plaintext for Ciphertext.
- Ciphertext Output:** Encrypted data block.

Decryption (Bottom)

- Identical Key Stream:** Crucially, the SAME key, nonce, and counter generate the same key stream.
- Recovering Plaintext:** The Key Stream is XORed with the incoming Ciphertext.
- Plaintext Output:** Decrypted data block.

Diagram detailing both Encryption and Decryption in Counter (CTR) Mode across multiple blocks.